HARMONIC ANALYSIS

- The Fourier series of a **known** function f(x) in a given interval may be found by finding the Fourier coefficients.
- The method described cannot be employed when f(x) is not known explicitly, but defined through the values of the function at some equidistant points
- In such a case, the integrals in Euler's formulae cannot be evaluated. Harmonic analysis is the process of finding the Fourier coefficients numerically.

To derive the relevant formulae for Fourier coefficients in Harmonic analysis, we employ the following result :

The mean value of a continuous function f(x) over the interval (a,b) denoted by **[f(x)]** is defined as $$[f(x)] = \frac{1}{b-a}\int_a^b f(x)dx .$$

The Fourier coefficients defined through Euler's formulae may be redefined as

$$a_0 = \left[\frac{1}{2l}\int_a^{a+2l} f(x)dx\right] = [f(x)]$$

$$a_n = 2\left[\frac{1}{2l}\int_a^{a+2l} f(x)\cos\left(\frac{n\pi x}{l}\right)dx\right] = 2\left[f(x)\cos\left(\frac{n\pi x}{l}\right)\right]$$

$$b_n = 2\left[\frac{1}{2l}\int_a^{a+2l} f(x)\sin\left(\frac{n\pi x}{l}\right)dx\right] = 2\left[f(x)\sin\left(\frac{n\pi x}{l}\right)\right]$$

Using these, we obtain the Fourier series of f(x). The term $a_1\cos x+b_1\sin x$ is called the first harmonic or fundamental harmonic, the term $a_2\cos 2x+b_2\sin 2x$ is called the second harmonic and so on. The amplitude of the first harmonic is $\sqrt{a_1^2+b_1^2}$ and that of second harmonic is $\sqrt{a_2^2+b_2^2}$ and so on.

Examples

1. Find the first two harmonics of the Fourier series of f(x) given the following table :

x	0	$\pi/3$	$2\pi/3$	π	$4\pi/3$	$5\pi/3$	2π
f(x)	1.0	1.4	1.9	1.7	1.5	1.2	1.0

Note that the values of $y = f(x)$ are spread over the interval $0 \le x \le 2\pi$ and $f(0) = f(2\pi) = 1.0$. Hence the function is periodic and so we omit the last value $f(2\pi) = 0$. We prepare the following table to compute the first two harmonics.

x^0	y = f(x)	cosx	cos2x	sinx	sin2x	ycosx	ycos2x	ysinx	ysin2x
0	1.0	1	1	0	0	1	1	0	0
60	1.4	0.5	-0.5	0.866	0.866	0.7	-0.7	1.2124	1.2124
120	1.9	-0.5	-0.5	0.866	-0.866	-0.95	-0.95	1.6454	-1.6454
180	1.7	-1	1	0	0	-1.7	1.7	0	0
240	1.5	-0.5	-0.5	-0.866	0.866	-0.75	-0.75	1.299	1.299
300	1.2	0.5	-0.5	-0.866	-0.866	0.6	-0.6	-1.0392	-1.0392
Total						-1.1	-0.3	3.1176	-0.1732

We have

$$a_n = 2\left[f(x)\cos\left(\frac{n\pi x}{l}\right)\right] = 2[y\cos nx]$$

$$b_n = 2\left[f(x)\sin\left(\frac{n\pi x}{l}\right)\right] = 2[y\sin nx]$$

as the length of interval= $2l = 2\pi$ or $l=\pi$

Putting, n=1,2, we get

$$a_1 = 2[y\cos x] = \frac{2\sum y\cos x}{6} = \frac{2(1.1)}{6} == -0.367$$

$$a_2 = 2[y\cos 2x] = \frac{2\sum y\cos 2x}{6} = \frac{2(-0.3)}{6} = -0.1$$

$$b_1 = [y\sin x] = \frac{2\sum y\sin x}{6} = 1.0392$$

$$b_2 = [y\sin 2x] = \frac{2\sum y\sin 2x}{6} = -0.0577$$

The first two harmonics are $a_1\cos x+b_1\sin x$ and $a_2\cos 2x+b_2\sin 2x$. That is $(-0.367\cos x + 1.0392\sin x)$ and $(-0.1\cos 2x - 0.0577\sin 2x)$

2. Express y as a Fourier series upto the third harmonic given the following values :

x	0	1	2	3	4	5
y	4	8	15	7	6	2

The values of y at x=0,1,2,3,4,5 are given and hence the interval of x should be $0 \le x < 6$. The length of the interval = 6-0 = 6, so that $2l = 6$ or $l = 3$.
The Fourier series upto the third harmonic is

$$y = a_0 + \left(a_1 \cos\frac{\pi x}{l} + b_1 \sin\frac{\pi x}{l} \right) + \left(a_2 \cos\frac{2\pi x}{l} + b_2 \sin\frac{2\pi x}{l} \right) + \left(a_3 \cos\frac{3\pi x}{l} + b_3 \sin\frac{3\pi x}{l} \right)$$

or

$$y = a_0 + \left(a_1 \cos\frac{\pi x}{3} + b_1 \sin\frac{\pi x}{3} \right) + \left(a_2 \cos\frac{2\pi x}{3} + b_2 \sin\frac{2\pi x}{3} \right) + \left(a_3 \cos\frac{3\pi x}{3} + b_3 \sin\frac{3\pi x}{3} \right)$$

Put $\theta = \frac{\pi x}{3}$, then

$$y = a_0 + (a_1 \cos\theta + b_1 \sin\theta) + (a_2 \cos 2\theta + b_2 \sin 2\theta) + (a_3 \cos 3\theta + b_3 \sin 3\theta) \qquad (1)$$

We prepare the following table using the given values :

x	$\theta = \frac{\pi x}{3}$	y	ycosθ	ycos2θ	ycos3θ	ysinθ	ysin2θ	ysin3θ
0	0	04	4	4	4	0	0	0
1	60°	08	4	-4	-8	6.928	6.928	0
2	120°	15	-7.5	-7.5	15	12.99	-12.99	0
3	180°	07	-7	7	-7	0	0	0
4	240°	06	-3	-3	6	-5.196	5.196	0
5	300°	02	1	-1	-2	-1.732	-1.732	0
Total		**42**	**-8.5**	**-4.5**	**8**	**12.99**	**-2.598**	**0**

$$a_0 = [f(x)] = [y] = \frac{\sum y}{6} = \frac{1}{6}(42) = 7$$

$$a_1 = 2[y\cos\theta] = \frac{2}{6}(-8.5) = -2.833$$

$$b_1 = 2[y\sin\theta] = \frac{2}{6}(12.99) = 4.33$$

$$a_2 = 2[y\cos 2\theta] = \frac{2}{6}(-4.5) = -1.5$$

$$b_2 = 2[y\sin 2\theta] = \frac{2}{6}(-2.598) = -0.866$$

$$a_3 = 2[y\cos 3\theta] = \frac{2}{6}(8) = 2.667$$

$$b_3 = 2[y\sin 3\theta] = 0$$

Using these in (1), we get

$$y = 7 - 2{,}833\cos\left(\frac{\pi x}{3}\right) + (4.33)\sin\left(\frac{\pi x}{3}\right) - 1.5\cos\left(\frac{2\pi x}{3}\right) - 0.866\sin\left(\frac{2\pi x}{3}\right) + 2.667\cos\pi x$$

This is the required Fourier series upto the third harmonic.

3. The following table gives the variations of a periodic current A over a period T :

t(secs)	0	T/6	T/3	T/2	2T/3	5T/6	T
A (amp)	1.98	1.30	1.05	1.30	-0.88	-0.25	1.98

Show that there is a constant part of 0.75amp. in the current A and obtain the amplitude of the first harmonic.

Note that the values of A at t=0 and t=T are the same. Hence A(t) is a periodic function of period T. Let us denote $\theta = \left(\frac{2\pi}{T}\right)t$. We have

$$a_0 = [A]$$

$$a_1 = 2\left[A\cos\left(\frac{2\pi}{T}\right)t\right] = 2[A\cos\theta] \qquad (1)$$

$$b_1 = 2\left[A\sin\left(\frac{2\pi}{T}\right)t\right] = 2[A\sin\theta]$$

We prepare the following table:

t	$\theta = \frac{2\pi t}{T}$	A	cosθ	sinθ	Acosθ	Asinθ
0	0	1.98	1	0	1.98	0
T/6	60^0	1.30	0.5	0.866	0.65	1.1258
T/3	120^0	1.05	-0.5	0.866	-0.525	0.9093
T/2	180^0	1.30	-1	0	-1.30	0
2T/3	240^0	-0.88	-0.5	-0.866	0.44	0.7621
5T/6	300^0	-0.25	0.5	-0.866	-0.125	0.2165
Total		**4.5**			**1.12**	**3.0137**

Using the values of the table in (1), we get

$$a_0 = \frac{\sum A}{6} = \frac{4.5}{6} = 0.75$$

$$a_1 = \frac{2\sum A\cos\theta}{6} = \frac{1.12}{3} = 0.3733$$

$$b_1 = \frac{2\sum A\sin\theta}{6} = \frac{3.0137}{3} = 1.0046$$

The Fourier expansion upto the first harmonic is

$$A = \frac{a_0}{2} + a_1\cos\left(\frac{2\pi t}{T}\right) + b_1\sin\left(\frac{2\pi t}{T}\right)$$

$$= 0.75 + 0.3733\cos\left(\frac{2\pi t}{T}\right) + 1.0046\sin\left(\frac{2\pi t}{T}\right)$$

The expression shows that A has a constant part 0.75 in it. Also the amplitude of the first harmonic is $\sqrt{a_1^2 + b_1^2} = 1.0717$.